Kausik Naskar

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GitHub — LinkedIn — DSA Profile

Summary

Computer Science undergraduate with a strong foundation in Data Structures and Algorithms, backend development, and database systems. Experienced in building scalable applications with efficient data handling and retrieval. Published researcher in IEEE with work on AI-driven agricultural disease detection, and a submitted paper on explainable lung cancer detection using Vision Transformers. Solved **877+** DSA problems across platforms and passionate about developing data-driven, AI-powered systems.

Education

Narula Institute of Technology 2023 – 2027
Bachelor of Technology in Computer Science
CGPA: 8.97 (till now)

Skills

Languages: Python, Java, C++, JavaScript, SQL
Web & Backend: React.js, Node.js, REST APIs, JWT Authentication
AI/ML & Data: LangChain, RAG Pipelines, Pandas, NumPy, TensorFlow
DBMS: PostgreSQL, MongoDB, Pinecone (Vector DB)
Tools: Git, Linux, Postman, VS Code, Anaconda, Jupyter, Vercel

Projects

AI Interview System

Full Stack + AI System — React, Node.js, PostgreSQL, REST APIs

- Developed an AI-powered interview platform to simulate real-world interview environments with dynamic question generation, supporting multiple interview modes (technical, behavioural, domain-specific).
- Designed normalised relational database schemas in PostgreSQL to store users, sessions, questions, and responses with referential integrity.
- Built RESTful backend APIs in Node.js for session lifecycle management, authentication, and real-time interaction between candidate and AI interviewer.
- Enabled automated performance evaluation by analysing stored response data, generating per-session scores and qualitative feedback reports.

Code RAG Assistant

RAG System — Python, Streamlit, Vector Database, LLM

- Developed a Retrieval-Augmented Generation (RAG) system that allows developers to query large codebases in natural language, significantly reducing time spent on code search and documentation lookup.
- Implemented embedding storage and semantic search using Pinecone vector database, enabling sub-second retrieval of contextually relevant code snippets from thousands of files.
- Designed end-to-end data pipelines for ingesting, chunking, and embedding both structured (modules,

functions) and unstructured (README, docstrings) code artefacts.

- Optimised retrieval accuracy through hybrid search strategies and prompt engineering, improving answer relevance by reducing hallucination on domain-specific code queries.

Software Engineering Team AI

Full Stack + AI Platform — React.js, Node.js, PostgreSQL, JWT, REST APIs, Vercel

- Built an AI-powered team collaboration platform for software development teams, enabling project management, task tracking, secure user authentication, and workflow automation.
- Designed relational database schemas in PostgreSQL to manage projects, tasks, teams, and user roles with referential integrity across the platform.
- Implemented **JWT-based authentication** and role-based access control to secure user sessions and protect sensitive project data.
- Developed scalable RESTful backend APIs in Node.js for real-time task updates, team collaboration, and workflow automation, improving overall team productivity and project execution.
- Deployed on **Vercel** with a responsive React.js frontend, ensuring smooth performance across devices.

Research Publications

- ***AI-Driven Crop Disease Prediction and Management System*** *Published — IEEE*
  Designed and trained a **ResNet50**-based convolutional neural network for crop disease detection, achieving **97.18% accuracy** across multiple plant species and disease classes. The system was integrated with a real-time web application that allows farmers to upload field photographs and receive instant disease diagnosis with treatment recommendations. The pipeline included custom data augmentation, transfer learning fine-tuning, and REST API deployment. **Published in IEEE**; available via the IEEE Digital Library.

- ***Vision Transformer-Based Explainable Framework for Automated Lung Cancer Detection*** *Under Review — SIGNASS*
  Developed a **Vision Transformer (ViT)** model for automated lung cancer classification on the **LC25000** histopathological image dataset, achieving **99.3% accuracy**. The framework incorporates explainability through attention-map visualisations (attention rollout), enabling clinicians to interpret model decisions at the patch level. Compared against CNN baselines (ResNet, EfficientNet) across precision, recall, F1-score, and AUC metrics. Currently under peer review at **SIGNASS**.

Achievements

- **1st Rank** in Hack-O-Nit 2025 (Inter-college Hackathon)
- **8th Rank** in Smart Bengal Hackathon 2025
- Solved **877+** Data Structures and Algorithms problems across competitive programming platforms — **Codolio Profile**